Predicting Shipment ETA and Delay Propagation

Aiqon AI internship take-home. Secondary tables, ablations and investigations are in `report/appendix.md`; every number here is copied from a file generated by the code.

> **Summary.** The system predicts final ETA, the probability of a delay over 48 h, and which stage caused it, updating at five checkpoints. On **synthetic** shipments it cuts ETA error from 24.2 h (best baseline) to 16.0 h and its explanations agree with ground truth 84% of the time once most stages are observed. On **real** AIS data, for the one stage that could be tested, a learned correction improves vessels' declared arrival ETA by about 1.5–2 h but barely beats a trivial baseline. The simulator's central assumption, that congestion lengthens port time, was not detected in real data.

1. Problem and definitions

Delay = actual − scheduled time at a milestone (hours). **ETA** = scheduled delivery + predicted final delay. A shipment is **badly delayed** if the final delay exceeds 48 h. Milestones: departure, port arrival, discharge, customs clearance, delivery. Predictions are made at five **checkpoints** (booked, departed, at port, discharged, customs cleared), each using only information available at that moment.

2. Data

No public dataset records scheduled and actual times for every stage of a shipment, so the main experiments use a **causal simulator** written for this project: 2,000 shipments on ten India→Gulf / Singapore / Europe / US routes. Delay propagates as $d_k = m_k \cdot d_{k-1} + own_k$: part of the upstream delay is absorbed by slack and new delay is added at each stage. The simulator includes congestion × terminal inefficiency, missed berth windows, customs office hours, inspections, dispatch cut-offs and missing timestamps. **Only the wind data is real** (ERA5); every other parameter is an assumption. Because each delay hour's cause is logged, the final delay decomposes exactly by stage, which gives ground truth for scoring explanations.

3. System

- **Data contract and as-of features.** Timelines are validated; a feature is missing until its checkpoint is reached, never a future value. Tests alter everything after a checkpoint and require predictions there to stay identical.
- **Chain model** (explicit propagation): one gradient-boosted model per stage predicts that stage's delay increment from the previous stage's delay and its own drivers, rolled forward to delivery. **Direct model:** one model over all checkpoints. They are blended with a weight per checkpoint.
- **Uncertainty:** split-conformal 80% intervals and a calibrated delay probability, each fitted on data separate from model training.
- **Explanations:** hours gained or lost per stage versus a typical shipment, the main driver per stage (TreeSHAP), and a plain-English sentence from a fixed template.

4. Results on synthetic data

Train on the earliest 60% of shipments, calibrate on 15%, test on the latest 25% (500 shipments). ETA mean absolute error in hours:

Method	Booked	Departed	At port	Discharged	Customs cleared	Overall
Schedule as ETA	52.0	52.0	52.0	52.0	52.0	52.0
Best baseline (last delay + median remaining)	29.0	29.2	25.8	21.9	15.1	24.2
Direct model	22.7	22.8	18.8	13.9	7.0	17.0

Chain model	27.1	26.7	17.2	12.1	4.3	17.5
Blend (this system)	**23.1**	**23.8**	**16.9**	**11.8**	**4.2**	**16.0**

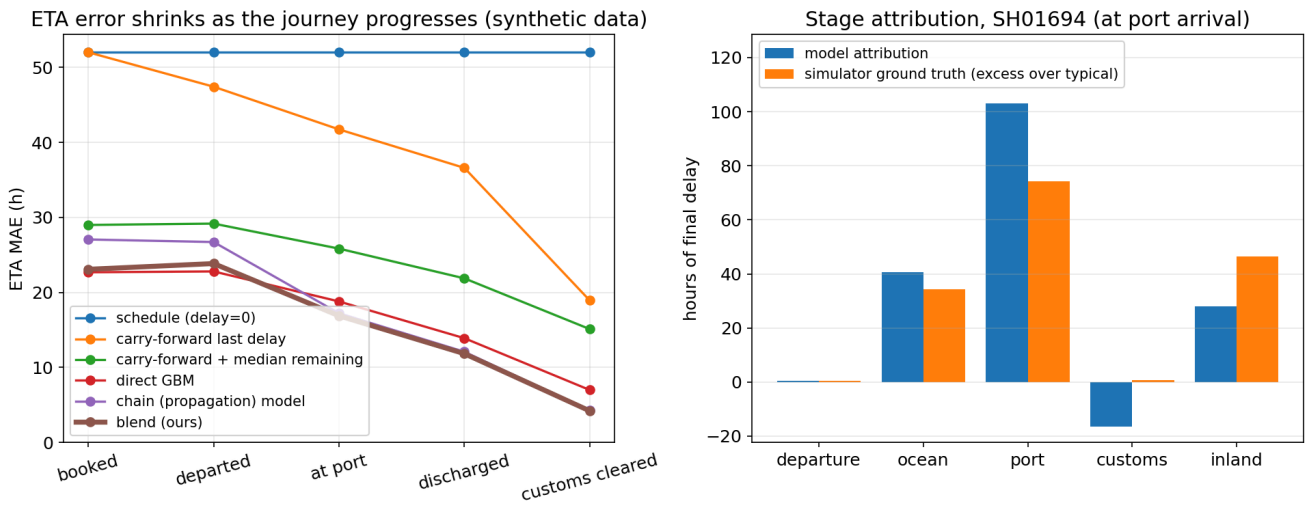


Figure 1 (synthetic data). Left: ETA error at each checkpoint. Right: hours of delay attributed to each stage for one held-out shipment, against the simulator's ground truth.

- **ETA.** Overall error is 16.0 h (95% CI 14.7–17.3) against 24.2 h for the best baseline, a paired improvement of 7.1–9.5 h. Error falls from 23.1 h at booking to 4.2 h after customs. Across three independently simulated worlds the blend averages 16.9 ± 1.1 h against 25.8 ± 2.0 h.
- **Uncertainty.** 80% intervals cover 0.77–0.87 of outcomes by checkpoint. The delay probability has AUC 0.89 overall (0.82 at booking, 0.99 after customs).
- **Explanations.** The stage blamed most agrees with ground truth 48% of the time at booking and 84% after customs (chance: 20%). They are weak early because the causes have not happened yet.
- **Weak spots.** On two routes never seen in training, error is 22.3 h (baseline 33.0 h) but interval coverage drops to 0.62–0.78. The blend beats the direct model alone by a small but reliable margin (0.7–1.5 h, 95% paired CI; 1.2–3.2 h on unseen routes), though not in one of three simulated worlds; the chain model is kept mainly because it supplies the stage-level explanations.

5. Evidence from real data

Port-arrival ETA. One stage could be tested on real data: 24 days of Danish Maritime Authority open AIS, where vessels broadcast their declared destination and ETA. Rows are vessel-hour states chosen using only information known at that hour, including vessels that never arrive; the split is by time. About 30,000 rows, 27 ports. Error in hours on the latest 20%:

Arrival radius	Declared ETA	Per-port median correction	Learned model	Model vs declared (95% CI)	Model vs median correction (95% CI)
5 km	11.46	7.97	7.41	+2.75 to +5.41	−0.96 to +2.14
10 km	7.27	6.50	5.59	+1.11 to +2.25	+0.45 to +1.37
15 km	5.97	5.32	4.26	+1.28 to +2.17	+0.76 to +1.41

The gain is real but modest and depends on how arrival is defined. Against a trivial per-port correction the model wins by at most about an hour, and at 5 km it is not distinguishable. Declared ETAs are already good in the typical case (median error about 1.5 h); the error comes from stale ETAs and no-shows. An earlier design that kept only vessels which did arrive showed a far larger gain and was **retracted**: its error scaled with the look-back window, a sign that the selection leaked the outcome.

Do the simulator's mechanisms hold? Using 1,639 real port stays, I tested four mechanisms written into the simulator (direction and rough size only; levels are not comparable):

Mechanism	Real data	Simulator	Verdict
Congestion lengthens the port stage	correlation −0.03 (CI −0.08 to +0.03)	+0.55	**Not detected**
Port-stage duration is heavy-tailed	mean / median 1.71	1.22	Supported; real tail heavier
Longer stays when starting Fri–Sun	1.07× (CI 0.96–1.18)	at customs only	Not detected
Arrival lateness has a late tail	11% over 6 h late, 3% over 6 h early	62% over 6 h late	Supported

The congestion proxy is crude and these ports are rarely full, so this does not prove congestion is irrelevant. It does mean the simulator's strongest assumption is not externally validated, and that part of the model's skill on synthetic data is the recovery of rules I wrote.

6. Limitations and open issues

- **Synthetic data.** The models partly learn the simulator's own functional forms. Absolute errors do not predict real-world accuracy, and nothing here validates customs, delivery or end-to-end ETA on real shipments.
- **Chain exposure bias.** Stage models are trained on the true previous-stage delay but receive a predicted one, whose error is nearly as large as the delay's own spread for later stages. Three corrections were tested on validation data (out-of-fold inputs, a sequential residual chain, a stacked combiner); none beat the current approach, so the bias is measured but not removed.
- **Shift.** Intervals lose coverage on unseen routes; the real-data model is far weaker on vessels it has not seen (8.67 vs 3.08 h).
- **Next step.** Linked milestones for the same shipment from a carrier or forwarder, to test the propagation model itself.

7. Reproduce

`pip install -r requirements.txt` (pinned, Python 3.13), then `python scripts/run_all.py` (about one minute) and `python -m pytest tests realworld -q` (24 tests). Runs are deterministic. Real-data steps and the audit log are in `realworld/README.md` and `report/appendix.md`.